

Assignment 3

Course: *Big Data*
Due date: *May 17th*

This is the second assignment related to your project. Work in pairs as you did in Assignment 3.

Perform exploratory data analysis on the Yellow Taxis dataset, enriched with the data you found in Assignment 3.

Use Pandas and Dask. Visualize your data. Your plots should be informative. Propose and calculate appropriate data aggregates (consider also temporal and spatial aggregations, analysis, and visualization).

Perform exploratory data analysis on the Yellow Taxis dataset, enriched with the data you found in Assignment 3.

Learn about DuckDB: <https://motherduck.com/blog/duckdb-tutorial-for-beginners/>.
Watch the DuckDB videos:

- DuckDB, an in-process analytical DBMS:
<https://www.youtube.com/watch?v=Z-6SnP6yzgo>
- Build a poor man's data lake from scratch with DuckDB:
<https://www.youtube.com/watch?v=33sxkrt6eYk>

Perform some of the tasks you performed with DuckDB with Dask SQL. Compare the performances.

Try using different data formats (parquet, csv) with different libraries (Dask, Pandas, DuckDB, Dask-SQL and compare the performance.

Be wise, try to find the most appropriate library for each task.